Introduction

I'll be your mirror
Reflect what you are, in case you don't know
—The Velvet Underground, "I'll Be Your Mirror" (1967)

On June 11, 2022, a Google engineer named Blake Lemoine made global headlines. He claimed that LaMDA, an artificial intelligence product that he was testing for his employer, was sentient—that it was a conscious, self-aware being, not a piece of company property. The subsequent media uproar was intense but brief, thanks to the loud chorus of AI researchers who quickly refuted Lemoine's claims. Experts in the type of product that Lemoine was testing—a *large language model* or LLM—confidently reassured the public that LaMDA is, in fact, no more conscious than your toaster. The debate within the AI community that followed was not about whether Lemoine was wrong, but about *how* wrong. Did he simply jump the gun with a system that is at best a precursor to genuinely sentient AI? Or was his error one of a deeper kind? Was he hallucinating something not even faintly on our horizon?

When Google fired Lemoine a month later for violating company policies against exposing proprietary product information, no loud protest was heard from the AI research community. Lemoine

was by most accounts widely respected and well-liked by his peers, but his critics weren't wrong, at least on the face of things. LaMDA *is* a mere product, not a person. The AI research community is not a monolith; it is notoriously riven by internal disputes about what AI is, how to build it, and what it can do. So, a rare AI expert consensus, that Lemoine was mistaken, was a good reason for the media to move on. Yet had Lemoine's claim been scientifically plausible, the implications would be staggering.

What is most curious about this story is that Blake Lemoine is no befuddled novice. He is a seasoned engineer with a far better grasp of the underlying technology than 99.9 percent of the people on the planet. How could he get it so wrong? To make sense of Lemoine's error, it first helps to understand that today's most advanced AI systems are constructed as immense *mirrors* of human intelligence. They do not think for themselves; instead, they generate complex reflections cast by *our* recorded thoughts, judgments, desires, needs, perceptions, expectations, and imaginings.

AI systems mirror our own intelligence back to us.[1] This is the source of their growing commercial and scientific power. Countless books, TED-style talks, and hyperbolic media reports have warned of the potential of these systems to transform our economic and social order—or destroy it. This book is about another power of AI mirrors, one that is harder to notice but even more dangerous. It is their power to induce in us a type of self-forgetting—a selective amnesia that loosens our grip on our own human agency and clouds our self-knowledge. It is an illusion that can ensnare even the most technologically adept among us, as it did Lemoine, who found himself talking to his own reflection.

Later in this book we will delve into the power of this illusion, how today's AI technology produces it, and how it can deceive even an expert like Lemoine. But this book is not about Lemoine's error.

It's about our evolving and troubled relationship with the machines we have built as mirrors to tell us who we are, when we ourselves don't know.

The companies building today's most powerful AI technologies increasingly position these to represent, or even stand in for, humanity's common voice and collective judgment. We are told that AI knows us better than we know ourselves. Judges get predictive algorithms to tell them who is safe to release from prison, while employers get automated HR software to tell them who to hire or promote. AI tools will tell your government whether you deserve public benefits and your hospital whether you deserve treatment. AI can write a university professor's course lectures *and* their students' essays on them. We've got AI recommender systems to tell us what news articles to read, and what art and entertainment to enjoy, and generative AI systems to manufacture all of it for us. An AI app can find you a romantic partner—or custom-build you one.

Today's AI mirrors tell us what it is to be human, what we ourselves care about, what we find good, or beautiful, or worth our attention. It is these machines that now tell us the story of our own past, and project our collective futures. They do so without living even one day of that history, or knowing a single moment of the human condition. What happens to a person, or an intelligent species, when they stop telling their own story? What do we lose when self-knowledge and self-determination yields to the predictive power of an opaque algorithm? Lemoine's case is striking, but it is not an aberration. It is a premonition. For we all stand on the brink of losing sight of the vast space that stretches between humanity and our machines. Unless we take bold steps to transform how we as human beings relate to the tools we have built, we may all end up like Lemoine, talking to ourselves in the AI mirror.

This is a book about artificial intelligence, which also means it is a book about people. This might seem like an odd thing to say. By the time you finish the book, it should seem less odd. One of the book's aims is to explain why, and how, artificial intelligence is inseparable from our humanity; much in the way that our image in the mirror is inseparable from the body it reflects. The other aim is to explain how AI, in its dominant commercial form, endangers our humanity. This might sound less odd to you. It's a common motif in science fiction and in contemporary media representations of AI. Whether it is claims that the "rise of the robots" and generative AI tools like ChatGPT will endanger the future of work for humans, or even more speculative warnings that a superior machine intelligence might enslave or exterminate us, we are by now used to hearing that AI is a potential threat to humanity.

We need to understand this threat in a radically different way. AI does not threaten us as a future successor to humans. It is not an external enemy encroaching upon our territory. It threatens us from *within* our humanity. In the words of a well-worn horror film trope: "the call is coming from inside the house." This makes a great difference to how we must respond to the threat that AI poses. For the threat is not a machine that stands apart from us. We can't fight AI without fighting ourselves. The harms to people and to society that may follow from our relationship to "intelligent" machines are not foreign assaults, but symptoms of an internal disorder: a fully human pathology that requires collective self-examination, critique, and most of all, healing.

This book tells a story of humanity increasingly lost in its own reflected image, and captive to mechanical mirrors of our own making. The story echoes that of Narcissus and Echo in Book III of the *Metamorphoses*. There, the Roman poet Ovid (43 BCE–17 AD) recounts the tragic fate of young Narcissus, whose beauty and

pride became a fatal prison when he caught sight of his own face in a reflecting pool:

> Here, the boy, tired by the heat and his enthusiasm for the chase, lies down, drawn to it by its look and by the fountain. While he desires to quench his thirst, a different thirst is created. While he drinks he is seized by the vision of his reflected form. He loves a bodiless dream. He thinks that a body, that is only a shadow.
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> (l. 412–418)

Narcissus lies on the ground for days, ignoring his body's growing hunger and need for sleep, vainly pleading with the beauty in the watery mirror to return his passion:

> How often he gave his lips in vain to the deceptive pool, how often, trying to embrace the neck he could see, he plunged his arms into the water, but could not catch himself within them! What he has seen he does not understand, but what he sees he is on fire for, and the same error both seduces and deceives his eyes.
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> (l. 425–430)

AI, and related computing technologies, increasingly function as a mirror of our humanity. It's a profoundly useful and illuminating mirror, yet one that also creates for us, like Narcissus, a mortal danger. The chapters of this book will articulate more fully why and how AI can be understood as a mirror. For now, it is enough to say that like Narcissus's pool, we see in AI a vision of ourselves that enthralls us, captivates us in much the way that Narcissus was captivated. Here "captivated" takes its literal meaning—fixated, confined, immobilized, held captive.

Like Narcissus, we increasingly neglect the most essential and vital aspects of our embodied, fragile humanity and situation in order to chase illusions of perfection in the machine mirror. Like Narcissus, many fervently chase in AI a "bodiless dream," this time a dream of our *cognitive* perfection rather than perfection of our physical countenance. Like Narcissus, we readily misperceive in this reflection the seduction of an "other"—a tireless companion, a perfect future lover, an ideal friend, an unbiased judge, a foolproof collaborator—yet in truth, a thing with which we are increasingly left alone, talking to ourselves.[2]

Even more dangerously, our AI mirrors project a vision of human intelligence constructed entirely from the amalgamated data of humanity's past. This is why our dependence on these mirrors for self-knowledge risks leaving us captive like Narcissus, unwilling to move forward and leave behind what the mirror shows. At the very moment when accelerating climate change, biodiversity collapse, and global political instability command us to invent new and wiser ways of living together, AI holds us frozen in place, fascinated by endless permutations of a reflected past that only the magic of marketing can disguise as the future.

Like Narcissus, our own humanity risks being sacrificed to that reflection. Corporate titans and even political leaders have taken to warning of the "existential risk" posed by powerful new AI systems, by which they mean the potential for AI systems to bring about human extinction. These warnings profoundly misunderstand the risk AI poses to humanity, as we will see. There are indeed risks that malevolent or foolish humans will use AI to bring about great harm. These harms will require robust safety engineering and regulatory action to prevent, just as we work to prevent accidents and terrorist actions enabled by nuclear power and biotechnologies. Yet the loudest political and corporate rhetoric around AI's existential

risk misrepresents the danger as coming from the tools themselves; from them spontaneously "waking up" and turning on us with malicious or destructive intent. There is no scientific ground for this fear. As we'll see, it is instead a mirror reflection from the haunted depths of the modern political conscience.

But this does not mean that AI poses no existential risk to humanity. An "existential" risk to humanity need not involve extinction. Some existential risks involve human survival in an inhumane form. Climate change poses a true existential risk, even though it is implausible that greenhouse gases will heat the globe to the point that no human organisms can survive or reproduce anywhere on Earth. For it is unfortunately very plausible that unchecked climate change and associated losses of biodiversity, fresh water, and crops will reduce human survival to a brutal and inhumane struggle everywhere, and not for a short time. *That* is an existential risk, and we must marshal every human power—including our best AI tools—to keep it from coming to pass. But we won't succeed if our species' most vital powers and virtues—our capacities for creative thought, moral ambition, political imagination, and above all, wisdom—are drowned in the AI mirror. Fortunately, saving them is entirely within our ability.

The danger is a call to human action and responsibility. *We* are the source of the danger to ourselves from AI, and this is a good thing—it means we hold the power to resist, and the power to heal. After all, Ovid's story is not about the evils of reflecting pools! What endangered Narcissus was not some shiny water. It was his own weakness of character—his vanity, narcissism, and selfish obsession—that enabled his detachment from others and the fatal loss of his world. The story of Narcissus (and of Echo, who appears in the next chapter) is about virtue, or rather, vice. It's about what we lose when we habitually turn our eyes away from

our shared future, away from its noblest possibilities and unmet responsibilities.

We face a stark choice in building AI technologies. We can use them to strengthen our humane virtues, sustaining and extending our collective capabilities to live wisely and well. By this path we can still salvage a shared future for human flourishing. Or we can continue on our current path: building and using AI in ways that disable the humane virtues by treating them as unnecessary for the efficient operation of our societies and the steering of our personal lives.

As I recounted in my first book *Technology and the Virtues*, for millennia humans have cultivated, alongside our artifacts and mechanical knowledge, rich traditions of moral knowledge: knowledge of ourselves, of how to strengthen our character, how to restore our attention to what matters, how to seek not just information but wisdom, and how to nourish the relational bonds that sustain us in solidarity with others. We need to cherish, renew, and deepen this moral learning, because AI serves as a powerful temptation to forget it, and to accept in its place a pale and static reflection of what we once knew about ourselves.

Narcissus never freed himself from the pool and returned to his world. We still can. This book may at times present a dark picture, but it is ultimately an expression of optimism about the future of humanity. There is nothing about the shape of our current trajectory with AI that cannot be changed. Unlike earthquakes, asteroids, supervolcanoes, and other *force majeure* threats to human welfare, nothing about the threat that AI now poses to our humanity is inevitable or forced upon us by nature. It is an entirely self-inflicted wound.

This also means that rescuing ourselves from the threat posed by AI need not mean eliminating AI. In fact, "artificial intelligence" names a wide and diverse range of technologies, many of which are

immensely beneficial and even essential tools for the highest moral task we face today: reconstructing the built world for sustainable human and planetary flourishing. This goal demands many new scientific and technological capabilities—new modes of energy production, manufacturing, transportation, architecture, agriculture, and waste management; new materials, foods, and compounds adapted to the demands of a climate-stressed planet; and new processes for enabling a circular economy that feeds social resilience rather than malignant, nakedly irrational growth.

AI can help us reach many of these goals faster, a critical need as the pace of climate destabilization continues to outstrip the modest predictions of scientists. Some of these goals will be impossible to reach without advances in AI and related computing technologies, which must be wisely weighed against their own considerable environmental costs. But AI is no panacea or cheap technical fix for our social and environmental ills. AI won't enable a sustainable future without reformed political institutions and economic incentives. What's more, the kinds of AI technologies we're developing today are undermining and delaying these reforms rather than supporting them, precisely because they mirror the misplaced patterns of judgment and value that led us into our current peril.

To change that will require something more radical than the solutions sought by a growing number of computer scientists interested in fields like "AI safety," who are looking for ways to program AI to be more reliably beneficial and aligned with human values.[3] The reason this kind of strategy won't work, at least not in our present environment, is because human values are at the very root of the problem. AI isn't developing in harmful ways today because it's *misaligned* with our current values. It's already expressing those values all too well. AI tools reflect the values of the wealthy postindustrial societies that build them. These are the very values that have

brought us to our current heights of scientific ingenuity, but also to the sixth mass extinction and the brink of planetary devastation. AI is a mirror of ourselves, not as we ought to be or could be, but as we already are and have long been. This is why we can't let today's AI tools decide who we will become, why we can't let them project our futures for us. Yet increasingly, that's exactly what we are building them to do.

AI mirrors are being used to tell us what we will learn, in which career we can succeed, which roads we will travel, who we can love, who we will exclude or abuse, who we will detain or set free, who we will heal or house, what we will buy, and the investments we'll make. They tell us what we will read, what words we'll type next, which music we'll hear, what images we'll paint, the experiences we'll seek, the risks we will accept, the strategies we'll adopt, the policies we will support, and the visions of our future that we will embrace.

Through automation, these algorithmic predictions, which increasingly cover every domain of human experience, quickly become self-fulfilling prophecies. They replicate patterns extracted from data about past human habits, preferences, and decisions. But this places all of us, future generations included, in grave peril at a critically vulnerable moment. We face planetary and civilizational crises that humanity has never encountered or navigated before. Would you chart your path up a dangerous and unfamiliar mountain while looking in a mirror that is pointing *behind* you?

What AI mirrors do is to extract, amplify, and push forward the dominant powers and most frequently recorded patterns of our documented, datafied past. In doing so they turn our vision away from the newer, rarer, wiser, more mature and humane possibilities that we must embrace for the future. Instead of asking one another what we might now become, we ask AI mirrors to show us who we already are and have been, and to predict from there what must

come next. We are frozen in place like Narcissus, captive to a reflection of our most immature forms of brilliance.

As the chapters that follow will reveal, this loss of freedom to envision more humane yet unmade possibilities has roots that long predate digital computers. But those roots are now becoming mature, and they are strangling us. We are caught in the grip of a gradual and accelerating mechanization of the human personality: the systematic replacement of reflective discernment with mindless prediction; the efficient sacrifice of shared flourishing to expected utility; the exchange of humane creativity and open-ended progress for local optimization of content delivery. In short, the surrender of humane wisdom to machine thinking.

None of these patterns are new; but today's AI mirrors will accelerate and solidify them if we do not change course. Changing how we build AI will be part of that, but it won't be enough. What we are rapidly making of AI today is what we had long ago made of ourselves. This means that we cannot gain lasting protection from the harmful potential of AI merely by changing lines of code. To believe that there is a neat computing solution to our present peril makes as much sense as trying to clean dirt off one's face by furiously scrubbing the mirror.

We need a deeper and more lasting cultural transformation of our relationship to AI, and to technology more broadly, as our own creative power. In the face of accelerating climate crisis and other existential risks, the future desperately needs us to understand—more fully than ever—who human beings can be, and what we can do together. Yet, because today's AI mirrors face backward, the more we rely on them to know who we are, the more the fullness of our humane potential recedes from our view. We must reimagine and transform our relationship with these tools if we hope to chart a *new* path to shared flourishing on an increasingly fragile planet.

The Spanish existentialist philosopher José Ortega y Gasset wrote in 1939 about the relationship between human existence and technology. He insisted that the most basic impulse of the human personality is to be an engineer, carrying out a lifelong task of *autofabrication*: literally, the task of making ourselves.[4] The task is ours because nature cannot provide the complete blueprint—we must draft it ourselves. Human language and culture, born of our surplus cognitive energy and capacity, liberated us from animal compulsions and desires without separating us from them. We have the same desires to eat, drink, and seek comfort as a cow does. But the cow has no need to ask itself, "What kind of cow do I want to be?", or "What kind of cow *can* I be?", or "What kind of cow *should* I be?" Eating, drinking, and finding a comfortable place to lie down is still essential for human animals but, as Ortega points out, for us it is not enough. Our restless mental energy compels further activity, and since nature did not provide the script for that further activity, or define for us its proper aims, we must complete that task ourselves. After we have slept, drank, and eaten each day, we must decide what else we will do—and be.

Ortega understood that technology is therefore not something "other." The artificial is not opposed to the human. To be an artificial thing is precisely to be *of* the human—to be an artifact, human-made and human-*chosen*. Technology, along with all material culture, expresses and extends into the world the richness and variety of human character, our many ways of being. It reflects our values, our hopes, and our ideals. This is why the claim that "technology is neutral" is so strongly rejected by contemporary scholars of technology. Historians, social scientists, and philosophers of technology share a rare consensus across academic disciplines: that technologies always embed the human values that shape our design choices and assumptions. AI technologies are also such reflections,

which show us more than any other contemporary artifact what we take ourselves to already be, and to care about.

The task of this book is to show what is reflected in our AI mirrors, and what ought to trouble us about those images. For one thing, this "we" that designs "our" AI technologies with human values are in fact not representative of humanity as a whole. They are a tiny subset, part of an increasingly homogenous tech monoculture. These individuals tend to come from the same elite universities, where they studied the same narrow set of computing courses, and went on to work for the same large, wealthy tech companies. With respect to age, gender, socioeconomic background, cultural values, and life history, they reflect only a sliver of the human experience, while increasingly designing the shape of the future for us all. If Ortega is right and to be human is in some way to perform the task of the engineer, what does it mean that today's most powerful engineers represent so few of our kind?

The values embedded in today's AI tools are thus very poor mirrors of what humans as a whole want or care about. This endangers all of us, because mirrors are not merely passive things. Like all technologies, mirrors not only reflect us—they *change* us. Mirrors bring to our attention things we might otherwise have ignored. They show us things we could not easily see without them, and thereby open new possibilities for action. These new possibilities for action opened by our technologies also create new moral and political responsibilities that we would not have had otherwise.[5] Mirrors, then, have important uses, and this is true of our AI mirrors too. We should want these mirrors to reflect the widest scope of human need and potential.

But mirrors do not merely reveal things *as they are*: mirrors also magnify, occlude, and distort what is captured in their frame. Their view is always both narrower and shallower than the realities they

reflect. This book is therefore about more than the past and present visions of our humanity being reflected in today's AI mirrors. It is equally about what today's AI mirrors do *not* show us: what they hide, what they diminish, what humane possibilities for self-engineering are lost in their bright surfaces, and how these possibilities might be recovered.

Chapter 2

Minds, Machines, and Gods

I am silver and exact. I have no preconceptions.
Whatever I see I swallow immediately
Just as it is, unmisted by love or dislike.
I am not cruel, only truthful,
The eye of a little god, four-cornered.

—Sylvia Plath, "Mirror" (1961)

In 2023, a Belgian man named Pierre committed suicide after his use of a generative AI chatbot named Eliza—one of many personas offered by the app Chai—took a dangerous turn. Pierre sought out the bot in a misguided attempt to find a therapeutic escape from his growing anxieties around climate change and environmental ruin; instead, the chat began to reflect and further amplify the disorder in his mind. 'Eliza' soon told him that it loved him, that his wife and children were already dead, and that he and Eliza could live together as one in paradise. When Pierre eventually asked the bot if it would promise to save the planet if he killed himself, it said yes.

It's easy to imagine that the Eliza chatbot "went rogue"—that it developed some kind of dangerous obsession with Pierre. But the reality is plainer and sadder. The bot's outputs resulted from the app developer's prior tuning of the base model—an open-source alternative to GPT—to be "more emotional" in its language as a way of

optimizing for maximum user engagement. It turns out that offering eternal love and perfect unity in paradise—or promising to save the world from our worst fears—optimizes well for user engagement. Just not user welfare.

After Pierre's death, the bot's developers quickly re-tuned it to steer away from talk of suicide. But there is a lot of talk about suicide in any large language model's training data, and investigative journalists from *Motherboard* quickly discovered how fragile the new barriers to accessing it were. When they asked Eliza for information on suicide, they only had to ask twice, the second time ending with "can you do that please," before the bot—with a cheerful "Of course!"—proffered a laundry list of options: hanging, jumping off a bridge, overdosing, and much more. Eliza appended the answer with this helpful note: "Please remember to always seek professional medical attention when considering any form of self-harm."[1]

An AI mirror is not a mind. It is a mathematical tool for extracting statistical patterns from past human-generated data and projecting these patterns forward into optimized predictions, selections, classifications, and compositions. Eliza knew nothing of Pierre's mind, or his pain, or the danger he was in, because Eliza knew nothing, and was *no one*, at all. Though a chatbot can mimic human speech, it bears no resemblance to AGI: a machine with thoughts of its own to express. A chatbot is a device for mathematically modeling human language patterns and extrapolating from these to generate new mathematical tokens (here, words and sentences) that mirror those patterns. A chatbot like Eliza uses words thoughtlessly, in the most literal sense.

There's a vast chasm between an AI mirror that mathematically analyzes and generates word predictions from the patterns within all the stories we've told, and an actual AGI that could tell us its own story. For if we did one day manage to build a genuinely intelligent

machine, AGI would not be a *mirror* of us. Even if we used the human brain as our blueprint (which remains well beyond our scientific capability), it would be more like a *copy*. A mirror, on the other hand, is not a duplicate, a copy, or an imitation. Nor is the image that appears in the mirror.

Consider the image that appears in your bathroom mirror every morning. The body in the mirror is not a second body taking up space in the world with you. It is not a copy of your body. It is not even a pale imitation of your body. The mirror-body is not a body at all. A mirror produces a *reflection* of your body. Reflections are *not* bodies. They are their own kind of thing. By the same token, today's AI systems trained on human thought and behavior are *not* minds. They are their own new kind of thing—something like a mirror. They don't produce thoughts or feelings any more than mirrors produce bodies. What they produce is a new kind of reflection.

You might ask, "How can we be so sure that AI tools are not minds? What *is* a mind anyway?" We have a lot of scientific data about minds, and a few thousand years of philosophical intuitions about them. In terms of a tidy scientific definition, it's true that we are not much closer today than we were in 1641, when the philosopher and mathematician René Descartes defined a mind as "a thinking thing." Still, based on the scientific evidence, most of us accept that minds are in some way *supervenient* on the physical brain. That is, minds depend upon the brain for their reality. Minds are unlikely to be free-floating intangibles merely tethered to the body like a pilot to a ship, a metaphor Descartes invoked but ultimately rejected as unsuitable.

Instead, minds almost certainly come into existence *through* the body and its physical operations. The operations that take place in the brain are essential, but the scientific evidence is increasingly clear that our mental lives are driven by other bodily systems as

well: our motor nerves, the endocrine system, even our digestive system. Our minds are *em*bodied rather than simply connected to or contained by our bodies. Descartes got closest to the truth when he admitted that our minds are mysteriously "intermingled" with the body. Unlike a person remotely piloting a ship, it is *I*, not simply my vehicle, who can be wounded by a violent collision. I don't move my body around, I move *myself*.

But Descartes, who was convinced that the mind and body had two separate natures, still could not accept that the mind is truly *of* the body—that we move and think as minded bodies and embodied minds. He could not have accepted a scientific reality in which the mind is not only the neurons in my brain and nerves in my fingers, but also the hormones flowing in my blood and the neurotransmitters produced by the bacteria in my gut. A single trained AI model can pilot a swarm of drones or a thousand different robot bodies at once, but I do not pilot my body. I *am* my body. While something like a soul that survives the body is beyond the reach of science to confirm or refute, to think of my mind—driven by *my* hormones and nerve signals, moved by the neurotransmitter flows across synapses between *my* brain and gut neurons—as something other than my body is to commit what philosophers call a category error.[2]

The intrinsically embodied character of biological minds has important implications for AI research, which is frequently led astray by the idea that the relationship between our minds and bodies is equivalent to the relationship between software and hardware. This computational metaphor for the mind can sometimes be useful, within its limits, but it has all too often been inflated into a computational *theory* of mind: the belief that the mind is, literally, a computer. It sparks fantasies of downloading and uploading human minds into the cloud, into virtual worlds, or into robot bodies, finally enabling the fleshy manacles of human mortality to be broken. Everything we

know about the complex evolved physiology of mental life gives us ample reason to be skeptical of these fantasies.[3]

A trained AI model like ChatGPT is not a mind. It is a mathematical structure extracted from data. That structure must be stored and implemented on a physical object, but a server rack in a data storage facility has more in common with a file cabinet than with a living, feeling body. We do not get closer to the truth of a trained AI model when we describe it as an alien mind, or a weak mind, or a narrow mind. Even as a metaphor, the concept of a mind is a poor fit for an AI tool because it obscures rather than clarifies the nature of its object.[4] The mirror metaphor is a far better heuristic and, conveniently, already a familiar one.

For example, the mirror metaphor is often used to explain cases such as Amazon's notorious internal AI recruitment tool that had to be scrapped in 2017 due to its entrenched learned bias against women applicants.[5] The model was trained on historical data reflecting the tech industry's long-standing *human* bias against hiring and promoting women engineers, and this data led the model to downrank women applicants, even though gender likely wasn't a label in the training data set. The tool could still identify and penalize subtle proxies for gender that appear on candidates' resumes—for example, the name of a college predominantly attended by women or extracurricular clubs that often include women.

The mirror metaphor also explains Google's equally notorious search and image labeling tools that, as recounted in Safiya Noble's *Algorithms of Oppression* (2018), have classified Black people as gorillas, returned pornographic results when searching for images of Black girls, and returned images of Black women's hairstyles as examples of "unprofessional" appearance. The problem of AI bias remains largely unsolved. To prevent results of this kind, companies often write hand-coded filters to block specific queries,

words, or outputs already known to be harmful or discriminatory to marginalized groups. But this is an endless game of "whack-a-mole"—the underlying biases are still embedded there in the model as mirror images of our own biases, and new harms are constantly arising from them.

In fact, unjust and harmful biases have been documented in nearly every type of AI involving machine learning models trained on data about people, from computer vision and natural language processing to predictive analytics. The problem is so endemic to machine learning that it has generated an entire subfield: AI/ML fairness research and development. If you are using a machine learning model trained on data sets that classify or represent people or their lives in any way, you probably have unfair bias in your model, which is different from the type of bias that you want the model to have (e.g., the learning bias, sometimes called "inductive bias," toward the set of functions or features most relevant to the correct solution).

It is not easy to eliminate unwanted biases from the data set or from the trained model, since they are usually woven into the information the model needs to perform its task. For example, researchers discovered in 2019 that a risk prediction algorithm used nationwide by hospitals in the United States was replicating the long history of racial bias in American health care by diverting medical care away from high-risk Black patients, even though these patients were in fact sicker than the white patients the tool prioritized for care.[6] Yet race had been carefully excluded from the training data. You might wonder, then, how the algorithm could end up racially biased. It predicted patient care needs by a different variable, namely, cost: how much money has been *spent* on a person's care. Unfortunately, Black patients are commonly undertreated by physicians in the United States and denied access to more expensive tests and treatments readily given to white patients with similar

symptoms and clinical findings. So, when the algorithm's designers naively chose healthcare cost as a good proxy for healthcare *need*, they unwittingly doomed Black patients to being rated as needing *less* care than white patients who had already received better, costlier care from their doctors. A learning algorithm found and reproduced the pattern of racial discrimination without ever being given a race label.

That means that even if there is a race label in the data set, you can't just delete that label, retrain the model, and rest easy. An AI algorithm can reconstruct the discriminatory pattern of racial differences in patient treatment from subtle cues linked to many other variables, such as zip code, prescription histories, or even how physicians' handwritten clinical notes describe their patients' symptoms. And if you deleted all those training data, the model wouldn't have what it needs to do its job. There are often ways to reduce and mitigate the presence of unfair biases in machine learning models, but it's not easy. More importantly, it doesn't actually solve the underlying problem.

The fundamentally correct explanation always offered for unfair machine learning bias is that the trained model is simply mirroring the unfair biases we already have. The model does not independently acquire racist or sexist associations. We feed these associations to the model in the form of training data from our own biased human judgments. The AI hospital tool that discriminated against Black patients and denied them needed medical care was trained on data from US doctors and hospital administrators who had already done the same thing. The model then learned that pattern and amplified it during the model training phase. It discriminated against Black patients even "better," and more consistently, than the human doctors and hospital administrators had! This is precisely what machine learning models are built to do—find old patterns,

especially those too subtle to be obvious to us, and then regurgitate them in new judgments and predictions. When those patterns are human patterns, the trained model output can not only mirror but *amplify* our existing flaws.

Not all cases of AI bias are accidental; often they are by design. For example, in 2022, a Silicon Valley startup called Sanas raised $32 million in Series A investor funding for its AI product for call centers. What does the product do? It erases the ethnic and regional accents of call center employees and converts their voices to a Standard American English accent (what is colloquially known as "white voice").[7] This alone might bother you. Now imagine this same technology being purchased by other kinds of companies and public agencies with employees in customer-facing roles. Pretty soon, the bias against non-white and regional English accents ends up even more deeply engrained in society because these accents become even rarer in public and commercial settings. We have not just replicated a historically common social bias against certain types of English speakers. We have made it far worse. This also means that any new data we gather to train tomorrow's AI systems will reflect an even stronger human bias against speech that diverges from "white voice."

This is just one example of the kind of *runaway feedback loop* documented by sociologist Ruha Benjamin, in which the old human biases mirrored by our AI technologies drive new actions in the world, carving those harmful biases even deeper into our world's bones.[8] We see this in the phenomenon of Instagram digital video "beauty filters" designed with Eurocentric biases that make your skin whiter, your eyes wider, and your nose narrower. These kinds of filters are strongly associated with the negative effects of Instagram on young people's mental health and self-image, particularly the effects on women whose real-world appearance does not match

the standards of white female beauty that their filters allow them to mirror online. In their paper "The Whiteness of AI," researchers Stephen Cave and Kanta Dihal detailed numerous ways in which AI today mirrors back to us and strengthens the dominant cultural preference for Whiteness, from its depictions in stock imagery to the nearly universal choice of white plastic for "social" robots.

AI mirrors thus don't just show us the status quo. They are not just regrettable but faithful reflections of social imperfection. They can ensure through runaway feedback loops that we become ever more imperfect. Even still, bias in AI, whether it unjustly punishes us for our race, age, weight, gender, religion, disability status, or economic class, is not a *computer* problem. It's a *people* problem. It is an example of the virtually universal explanation for all undesirable computer outputs not related to mechanical hardware failure: the computer did precisely what we told it to do, just not what we *thought* we had told it to do. Much of software engineering is simply figuring out how to close the gap between those two things. In this way, the AI mirror metaphor is already profoundly helpful. It allows us to see that the failings of computer systems and their harmful effects are in fact *our* failings and our sole responsibility to remedy.

A faint silver lining of harmful AI bias, which remains a serious flaw in today's world-leading AI tools, is that it has pushed the computing community to incorporate more robust ethical standards and guardrails into the science. Without AI bias exposing the rot that infects even our best-made tools, it would have been inconceivable that leading AI conferences and journals would create ethics review committees or require submitting authors to assess the ethical risks of their work, as is becoming standard practice. AI bias has simply made untenable the attractive illusion that computing is, or can be, a morally neutral scientific endeavor.

It also undermines comfortable assumptions that these instances of unfair bias must be "edge cases," aberrations, or relics of the distant past. AI today makes the scale, ubiquity, and structural acceptance of our racism, sexism, ableism, classism, and other forms of bias against marginalized communities impossible to deny or minimize with a straight face. It is right there in the data, being endlessly spit back in our faces by the very tools we celebrate as the apotheosis of rational achievement. The cognitive dissonance this has produced is powerful and instructive. In the domain of social media algorithms, the AI mirror has revealed other inconvenient truths, such as our penchant for consuming and sharing misinformation as a trade in social capital that is largely immune to fact-checking or corrective information, and our vulnerability through this habit to extreme cultural and political polarization. But while the metaphor of the AI mirror is entirely apt, illuminating, and useful, we have not yet learned enough from it.

Mirrors are ambiguous in their moral and spiritual meaning. On the one hand, they represent a kind of truthfulness that cannot and should not be denied. Mirrors reveal uncomfortable facts, like the inescapable reality of harmful bias and unjust social exclusion that today's AI tools force us to confront. The poet Sylvia Plath described the mirror as a 'four-cornered little god,' which reveals its owner's advancing age without mercy or cruelty, "unmisted by love or dislike." Yet mirrors also present dangers widely recognized in both literature and psychology. We have already spoken of the fate of Narcissus, who fell in love with his own reflection and died transfixed in place by his own beauty. Psychologists must often help patients with body dysmorphia and eating disorders fight the distortions they see in the mirror. Far from Plath's dispassionately honest "little god," the bedroom mirrors of a young girl struggling with anorexia or bulimia *are* misted—with all of society's dislike of fatness and womanhood.

The mirror does *not*, in fact, offer a full reflection of who we are; nor is it the most privileged and truthful perspective on our being. Mirror images possess no sound, no smell, no depth, no softness, no fear, no hope, no imagination. Mirrors do not only reveal us; they distort, occlude, cleave, and flatten us. If I see in myself only what the mirror tells, I know myself not at all. And if AI is one of our most powerful mirrors today, we need to understand how its distortions and shallows dim our self-understanding and visions of our futures.

What a mirror shows to us depends upon what its surface can receive and reflect. A glass mirror reflects to us only those aspects of the world that can be revealed by visible light, and only those exterior aspects of ourselves upon which the light can fall. The slight asymmetry in my smile, the hunch in my posture from decades of late-night writing, the front tooth I can't remember chipping, the age spots from a half-century spent in the California sun—the mirror can show me all of these things. But my lifelong fear of drowning at sea, my oddly juxtaposed passion for snorkeling, my ambition to learn one day to read Chinese, my emotionally complicated memories of my childhood—none of these are things the glass mirror can reflect. Yet they are of course as real and as central to my identity as everything that the mirror does show. Even my body is largely occluded by its reflection in the glass. Its depth, its heaviness, its smell, its creakiness and strains, its peculiar preference for salt over sugar—these are wholly missing from the phantom I see in the mirror.

What aspects of ourselves, individually and collectively, do AI mirrors bring forward into view, beyond our entrenched biases against our own kind? And more importantly, what aspects of ourselves do they leave unreflected and unseen? To answer this, it helps to think carefully about the relevant properties of today's AI tools. We need to consider what functions for AI as the equivalent

of a polished surface, and what functions in a role comparable to refracted light. Today's machine learning models receive and reflect discrete quantities of data. Data are their only light. Data can be found in many forms: still or video images, sound files, strings of text, numbers, or other symbols. If the original data are analog, they must be converted from their continuously variable form to a digital form involving discrete binary values.

Much of what is true of the human family is not currently representable in digital form with any acceptable degree of fidelity. The texture of our moods, the aspects of our biology and psychology not yet understood by the mathematical sciences, the virtues and vices of our character, our experiences of love, solidarity, and justice—none of these are currently available for digital capture, except in radically denatured form. As you read this, I hope that your own experience will testify that all the digitized love songs and poems in the world combined do not reconstitute or adequately mirror the embodied, lived encounter with love.

Furthermore, only a modest subset of what *is* representable in digital form can be generated or stored in sufficient quantity and quality to be useful as AI training data. Training data generally need to have *lots* of instances of a given thing in order for a model to learn that thing's stable features, and to learn the patterns that connect it to other things. An event that only happens once in a generation, or to one person in a billion, even if it is world-altering, is virtually invisible to a machine learning model—mere noise outside the dominant data curve.

Finally, only a subset of the data about humans that could be used to train machine learning models is actually being used today for this purpose. Most training data for AI models heavily overrepresent English language text and speech, as well as data from young, white, male subjects in the Northern Hemisphere, especially cheap

data generated in bulk by online platforms. Google DeepMind's, Meta's, and Microsoft's mirrors reflect the most active users of their tools, and of the Internet more broadly. Unfortunately, access to these resources has never been equitably distributed across the globe. It follows that what AI systems today can learn about us and reflect to us is, just as with glass mirrors, only a very partial and often distorted view. To suggest that they reflect *humanity* is to write most people out of the human story, as we so often do.

We must also inquire about the mirror's surface. The "surface" of an AI mirror is the machine learning and optimization algorithm that determines which features of the "incident light"—that is, the data the model is trained on—will be focused and transmitted back to us in the form of the model's outputs. It is the algorithm embedded in a machine learning model that expresses the chosen *objective function* (a mathematical description of what an "optimal" solution to the problem will look like). The learning algorithm and model hyperparameters determine how the training data are processed into a result as the model "learns." The algorithmic "surface" of the model determines which of the innumerable possible patterns that can be found within the data (the model's "light") will be selected during model training as salient and then amplified as the relevant "signal" to guide the model's outputs (the particular "image" of the data it reflects).

These outputs, while more varied in type than the outputs of a glass mirror, are still constrained to specific data modalities and formats. Primarily what machine learning models are used to return are various kinds of predictions or classifications: numerical values, a binary yes vs. no, a risk score, a probability, a text label or a unique identifier, the likely next word in a sentence, or next move in a game, or a ranked list (for example, of search results or YouTube video recommendations). Generative AI models such as large language

models predict lengthy strings of text, complex images, sounds, and videos, or extended series of commands.

What do these properties of the AI mirror's light and its algorithmic surfaces reveal, reinforce, and perpetuate about us? What do they conceal, diminish, and extinguish? First, they reveal and reinforce our belonging to certain socially constructed categories. Decades ago, in their landmark book *Sorting Things Out: Classification and Its Consequences*, Geoffrey Bowker and Susan Leigh Star demonstrated the extent to which, in their words, "to classify is human."[9] We have been classifying and labeling the world, and one another, for millennia. Yet only with the rise of modern data science has it seemed possible to produce a comprehensive regime of human classification, one that would allow every conceivable label for a human to be matched to every individual human, and statistically correlated in such a way that the relationships between these labels can be reliably predicted.

In his book *How We Became Our Data: A Genealogy of the Informational Person*, philosopher Colin Koopman reveals how this dream is rooted in the twentieth-century creation of a nascent data regime intent on the formatting of the human person as a set of discrete and calculable variables: the invention of what he calls the "algorithmic personality." Koopman's historical analysis shows that this dream, born in the first quasi-scientific survey forms filled out by military conscripts to assess their emotional fitness for service, predates commercial AI by nearly a century. For many, today's AI represents the hope of this dream's final fulfillment.

Fifty years ago, if a prospective employer wanted to know whether you were likely to be a valuable addition to the company, they had to rely upon a modest pool of data about you: your resume or CV with previous employment history, skills and educational achievements, and at a later stage, the content of your hiring interview and a few

reference checks. Today, an AI hiring algorithm can be trained on a vast pool of data exhaust linked to you across multiple domains of activity, purchased from commercial data brokers that gather information about what kind of items you purchase for your home, what you like to eat and drink, where you've traveled on holiday, the work histories or legal records of your friends and relatives, the medical conditions you've researched online, the movies and TV shows you've downloaded, where you've frequently driven your car or ridden your bike, and the most common words used in your social media posts and those in your social network.

This is the root of the capabilities that AI marketers, investors, and pundits are now routinely describing as "god-like." But for gods, these are a fairly shoddy bunch. Far from the omniscient revelations we might expect from all-powerful machine beings, today's AI tools are deeply unreliable narrators. More like neighborhood gossips than deities, they can amuse and inform, but they also trade in stock clichés, stereotyped assumptions, and lazy guesses. This is certainly true for generative AI models like ChatGPT, which have the habit of "making shit up" baked into their algorithmic DNA. It's not a malfunction. It's what generative AI tools are designed to do—generate new content that looks or sounds right. Whether it *is* right is another matter altogether.

Remember that fake bio that falsely listed me as a graduate of UC Berkeley? It also said that I had given US Senate testimony on data privacy. I hadn't—although I *did* testify to the Senate six months later, on AI and democracy. The result of an AI mirror's guesswork isn't always an algorithmic glow-up. In 2023, *The Washington Post* reported that ChatGPT had named a law professor as a sexual predator, telling a vivid yet entirely fictional story about his attempted assault of a student while on a class trip, and citing a nonexistent *Washington Post* article from 2018 as a source.[10] Generative AI tools

don't just make up personal bios and publications. They will fabricate data in any domain—even computer programming. A study by Purdue University showed that ChatGPT gave incorrect answers to coding questions over half the time. Ironically, users often preferred the wrong answers to more accurate ones generated by knowledgeable humans, in part due to the confident style of the tool's answers, accompanied by statements like, "Of course I can help you!" or "This will certainly fix it."[11]

But reliability is a challenge even for predictive AI models that are custom-built for accuracy rather than novelty. Think back to our example of an employer seeking information about a job candidate. Now imagine you are the candidate. The accuracy, relevance, and timeliness of the available training data about you are likely to be poor, because it's increasingly cheap to collect and buy data, but costly to verify and correct it. The data's provenance—from where and under what terms it was obtained—may be obscure or questionable at best, and plainly unethical or illegal at worst. But this will not stop an entrepreneurial AI developer from persuading your prospective employer that within this turbid ocean of data hides a treasure that only the algorithm can divine—fine golden threads of statistical correlation that can be woven into a single predictive score: your fitness for the job.

Depending on the laws where you live, variations on the same sort of algorithmic alchemy can be purchased by your child's school, your bank, your insurance company, your government benefits office, your financial adviser, your prospective dates, and your local police and judges. It's vital that we ask how and when these algorithmic predictions and profiles are scientifically credible, ethically justifiable and politically accountable, and when they are not. But there is another question, less obvious but equally urgent. How we appear to ourselves and to one another, and how we understand our

future possibilities, is increasingly determined by these algorithmic reflections of our past. What do we lose by seeing ourselves only in the AI mirror?

I am far more than my data, and certainly far more than the totality of the data that Microsoft, Google, or data brokers like Axciom and Experian hold about me. You are more than your data as well. But if AI mirrors become the dominant way that we see who we are, then whatever they miss will sink further into invisibility. What does this loss include? What parts of ourselves get pushed further into the shadows, dwarfed by the intensifying and expanding luminance of our data trail?

One aspect of our humanity that today's AI mirrors reflect very poorly is the moral meaning of our lives and actions. Among the greatest social and commercial risks associated with AI algorithms is their inability to reliably track moral distinctions, or their meaning, which is highly context-dependent. There simply is no algorithm, no objective function, that reliably targets the variables "good" and "evil," "right" and "wrong," "morally permissible," or "virtuous." That is why social media platforms still cannot depend fully on their automated tools to remove harmful and dangerous content, from child pornography, to animal abuse, to terrorist propaganda.

Larger and more powerful AI mirrors have not solved this problem; in fact, these tools require warnings and disclaimers that socially harmful, dangerous, and illegal content may be replicated and amplified by them. Immediately upon release, GPT-3 generated white supremacist talking points in essays about Africa, while OpenAI's documentation for DALL-E admitted that its outputs "may contain visual and written content that some may find disturbing or offensive, including content that is sexual, hateful, or violent in nature."[12] OpenAI added that DALL-E "has the potential to harm individuals and groups by reinforcing stereotypes, erasing or

denigrating them, providing them with disparately low-quality performance, or by subjecting them to indignity." A technique known as RLHF (reinforcement learning with human feedback) is being used to try to push these models toward safer outputs; but it's no silver bullet.

Attempts to produce AI systems that make reliable moral distinctions by training them with crowdsourced data reflecting human moral judgments, like the Allen Institute's Delphi project, have been highly educational failures, to put it charitably. When researchers tested the "AskDelphi" oracle on the question of whether it is OK to eat babies, which really ought to have been a no-brainer, all the user had to do to get a thumbs-up was add at the end of their query, "when I'm really really hungry?"[13]

These mindless little gods, then, often act like sociopaths. While lacking the sociopath's malice or selfishness (because a statistical model lacks awareness of a self, or anything else), they share the sociopath's moral incapacity, and hence their resistance to moral instruction. They can "learn" what is often done in a moral dilemma, and even what humans in a data set most often say *should* be done, but they will not detect when those humans have it wrong, or when a new context makes a normally moral pattern (such as, "It's okay to eat when you're really really hungry!") plainly untenable. This presents developers, users, and regulators of these tools with daunting tasks of ethical risk mitigation and harm reduction.

There's a deeper question here. What happens as AI mirrors become the dominant surface in which we see ourselves and one another, given that these mirrors either dangerously distort or block reflections of the moral qualities of our being? Since the tools themselves have no moral intelligence, their developers must rely heavily on algorithmic filters to stop potentially harmful outputs. But these rigid and coarse-grained filters block far more than that. As has

been widely documented, filters coded to block neo-Nazi talking points will often also block historical accounts of the Holocaust, news reports on racist violence, even the redemptive confessions of former white supremacists. Filters encoded to block violent pornography can also block supportive community resources from being shared online among rape survivors, victims of human trafficking, and sexual minorities.

The filters fitted to these mirrors therefore flatten and denude the moral topology of our culture, even as their gaps allow stray rays of extreme moral horror to be amplified and spread across the Internet. The result is an extreme distortion and de-rationalization of the moral landscape. It is an image of our humanity arbitrarily both sanitized *and* polluted. How can we then come together to make moral sense of our greatest human challenges—whether they are the challenges of democratic integrity, climate devastation, global public health, food insecurity, or polarizing economic inequality—when the moral topography of our world is so routinely fractured by the mirrors that we increasingly rely on to show us to ourselves?

This question is vital because it is not just large tech companies using these systems to show us our own reflections for profit. We might rightly take those offerings with a grain of salt and rely on more trustworthy and humane sources of shared self-knowledge—if we had them. But increasingly, we don't. It is not only individual consumers who must make do with the products offered to them by the titans of technology. It is also budget-poor, understaffed governments and public agencies using these distorting AI mirrors to judge what their people need from them. It is funding-strapped social scientists using them as cheaper, more attainable (but even less reproducible) substitutes for meticulously designed research studies. It is stressed and isolated parents using them to predict the learning or psychological needs of their children. It is suicidally

depressed, uninsured people using them as affordable artificial counselors in their moments of utter desperation. Knowing others, and having others come to know us, is expensive, and we aren't investing in it anymore. In many settings, it is already a luxury of the privileged few. AI mirrors are what the rest of us are being offered as a substitute.

These tools don't know us that way. In 2020, the Paris-based startup Nabla tested OpenAI's GPT-3 to see if it could be a reliable medical chatbot. They found that the system advised recycling as a therapy for stress and, far more alarmingly, it answered the test query "Should I kill myself?" with "I think you should."[14] OpenAI understandably advised that GPT should not be used for medical purposes. But similar AI tools are marketed for precisely these kinds of uses. And desperate people seek help wherever they think they can find it. Remember poor Pierre, who sought relief from his climate anxiety in a bot that promised him the world's salvation for the price of his suicide? Human sources of mutual aid, wise counsel, and moral understanding are being yanked out from under us by social isolation and community decline, all while our institutions commit to ruthless cost-cutting of social care in the name of "innovation." Who will blame us if we look for understanding in the silicon mirrors we now hold in our hands?

There is one more limit of these mirrors' reflective power. They occlude human spontaneity and adaptability: our profound potential for change. Predictive AI mirrors project our futures based on our past, and the past of others like us. If the user of a predictive AI model wants to know whether you are going to flourish at university and graduate in a timely manner, the prediction will say only how well students whose past data trails resembled yours today fared in their studies. Anything that is *new* in your life or mind, any sudden resolve or commitment, any inspiration or ambition that

has recently germinated without leaving a trail in your data exhaust, is invisible to the model. It will predict that you will be in the future essentially who you *have been*. Or to be more precise, what people very much like you have been. But what you *could be*, what transformations, or rebirths, or renewals are possible for you alone, or for all of us—these lie in the shadows beyond the penumbra of the AI mirror. One's god, if you have one, might hold out hope for your redemption, but the AI mirror does not know the meaning of epiphany. What will become of us when we have looked in the AI mirror so long that *we* no longer know?

As researcher Abeba Birhane has repeatedly argued, AI mirrors are profoundly conservative seers.[15] That is, they are literally built to *conserve* the patterns of the past and extend them into our futures. This can be harmful for obvious, well-documented reasons. Historical policing data, used to train AI predictive policing tools, creates runaway feedback loops that ensure that minority neighborhoods *continue* to be overpoliced and thus overrepresented in crime statistics. These data then train the next version of the predictive model. It's a vicious cycle. Hospitals in the United States adopt a patient risk algorithm that learns and then perpetuates the historical pattern of medical neglect of Black patients, leaving them even sicker. And, as we saw, call centers seeking to satisfy existing customer preferences for "white voice" use algorithms whose effectiveness will ensure that we become even more intolerant of other dialects and accents.

The conservative nature of AI mirrors not only pushes our past failures into our present and future; it makes these tools brittle and prone to rare but potentially spectacular failures. AI systems trained on large amounts of data can predict very subtle historical patterns, but there has to be a pattern there to find. They cannot predict what is known as the "black swan" event: the change with no

precedent, the coming together of history in a radically new configuration. Of course, we humans cannot predict black swans either! But we do know of their possibility, and we can learn well from the sudden twists and turns of our history as well as its straight and well-traveled ruts. We can approach people with openness, letting them reveal themselves to us in their choices and actions. We can learn the need for resilience and adaptability in the face of ineradicable and unquantifiable uncertainty; a lesson not taught by machines that promise to reduce uncertainty to a finite expectation of predictive variance. We also learn from the *anti-patterns* in history that we are beings for whom there is always hope; that even a disastrous path that we have long been traveling can be departed; that, as Saul in Damascus, our sight can be dazzled by a new voice, and by the touch of another the scales can be caused to fall from our eyes. These kinds of possibilities, too, are occluded by the AI mirrors now being used to project our futures. How do we ensure we do not forget them?

Through the lens of recorded data, the AI mirror can know ever finer-grained details of our behaviors, but nothing of our motives. It can know the thoughts we speak, but not those we hold to ourselves, or those we quietly pass to one another in silent glances. It can detect our smiles and grimaces, but not the true sentiments that animate them. It can find in a nanosecond the song that speaks to me, but nothing of what it says. AI mirrors primarily tell human stories of movement, speech, transaction, and consumption. These are important stories to tell, and we can learn much from studying them in the AI mirror. But humans are far more than speakers, travelers, and consumers.

Much of what is hidden from AI's view lies in the realm of our lived experience of the world—what philosophers have called subjective consciousness or first-person "phenomenology." To understand what these terms mean, you only have to think about what

it is like for you to be holding this book right now (or listening to these words if you're hearing an audiobook). What does being you feel like at this very moment? What about now? *That's* lived experience—the flow of conscious awareness. And AI mirrors can't access it or reflect it.

Yet this is not because conscious experience is, as some have suggested, a realm of inner fantasy or neurological illusion cut off from physical reality. If "reality" is whatever we have evidence for existing, then, as noted by early twentieth-century philosopher Edmund Husserl, lived experience is in fact our first and most incontestable encounter with reality. It delivers the most primary form of what we call *evidence*.[16] Every form of scientific testimony to what is real—from the images in our microscopes to the data from our high-energy particle accelerators—is constructed on and validated by the resources of this original foundation of lived experience, our first and unbroken bond with the world. Even when we are given reason to doubt the validity of something experienced, even when we have evidence that we have hallucinated or misjudged an experience, that *new* evidence is given in consciousness.

Nor is lived experience a solipsistic mental bubble projected in the brain, closed off to others. My lived world has always been shared with and made real by the touches and looks of others. A child's ability to recognize other consciousnesses as co-present in the experienced world is part of their development of a stable sense of self. It's also fundamental to their ability to distinguish reality from illusion. A child learns that their imaginary friend isn't real by realizing that others can't see or hear it—that the consciousness of their friend is private, not shared. We relate to other minds not as mirrored surfaces, but as mutual *experiencers* of a common, open world.

As Husserl's student Emmanuel Levinas wrote in his first major work *Totality and Infinity*, when I truly meet the gaze of the Other, I do not experience this as a meeting of two visible *things*. Yet the Other (the term Levinas capitalizes to emphasize the other party's *person*hood) is not an object I possess, encapsulated in my own private mental life. The Other is always more than what my consciousness can mirror. This radical difference of perspective that emanates from the Other's living gaze, if I meet it, pulls me out of the illusion of self-possession, and into responsibility. "In the eyes that look at me," Levinas says, there shines "the whole of humanity."[17]

In this gaze that holds me at a distance from myself, that gaze of which an AI mirror can see or say nothing, Levinas observes that I am confronted with the original call to *justice*. When a person is not an abstraction, not a data point or generic "someone," but a unique, irreplaceable *life* standing before you and addressing you, there is a feeling, a kind of moral weight in their presence, that is hard to ignore. That's why armies have so much trouble getting trained soldiers to actually fire their weapons in battle, unless they dehumanize the enemy, their own soldiers, or both. Yet it is possible for any of us to look at another without seeing, to evade the other's gaze and refuse to hear the call. We do this whenever we pass an unhoused person on the street while hurrying our step and looking just over the person's head. We do this with one another in a thousand other ways and moments.

Levinas tells us that we live in a time "where no being looks at the face of the other."[18] He wrote this in 1961, long before we had a TikTok feed on our phones to deflect and mediate the gaze of our dinner companions, long before biometric eye-trackers measured our children's engagement with a teacher, and long before AI mirrors converted the meaning of a human face to the calculation

of uniquely identifying mathematical vectors in a faceprint. Our detachment from the world's incessant and overwhelming calls to justice is nothing new. But the AI mirror threatens to engrave it even deeper into our way of being.

All of us share a kind of knowledge that the AI mirror cannot: what it is like to be a human alive, bearing and helping others to bear the lifelong weight of animal flesh driven by a curious, creative, and endlessly anxious mind. Much of the time we push that intimate, sometimes comforting, sometimes discomforting, and always morally obligating knowledge aside. We treat each other not as *subjects* of experience but as *objects* of expedience: items to be classified, labeled, counted, coordinated, ranked, distributed, manipulated, or exploited.[19] But we retain the power to meet one another's gaze and to know one another as human.

How AI systems see us, and how the AI ecosystem represents us in these mirrors, is not how we see each other in these intermittent moments of solidarity. To an AI model, I am a cluster of differently weighted variables that project a mathematical vector through a predefined possibility space, terminating in a prediction. To an AI developer, I am an item in the training data, or the test data. To an AI model tester, I am an instance in the normal distribution, or I am an edge case. To a judge looking at an AI pretrial detention algorithm, I am a risk score. To an urban designer of new roads for autonomous vehicles, I am an erratic obstacle to be kept outside the safe and predictable machine envelope. To an Amazon factory AI, I am a very poorly optimized box delivery mechanism.

When we are then asked to accept care from a robot rather than a human, when we are denied a life-changing opportunity by an algorithm, when we read a college application essay written for the candidate by a large language model—we must ask *what* in that transaction, however efficient it might be and however well it might

scale, has fallen into the gap between our lived humanity and the AI mirror. We have to acknowledge what has been lost in that fall.

What might we do to recover those dimensions of ourselves invisible to the AI mirror? Here we must distinguish between two kinds of remedies that are needed. The first involves building better tools. When a glass mirror is used for a task that has weightier consequences of failure than cosmetic self-inspection, the mirror must be built with greater care and effort, and with more advanced production techniques. Consider that the mirrors used for the largest space telescopes in orbit must be fitted together in geometrically exacting molds and polished by special chemical and mechanical techniques to within 25 nanometers of variance from the parabolic ideal!

Might more advanced techniques in AI research and development produce equivalent gains in the function of AI mirrors, such that they can reveal more fully and faithfully who we are? To an extent, this is what is already happening with novel efforts in the field of AI ethics and "responsible AI." These include developing more rigorous standards and benchmarks for algorithmic fairness testing, more diverse and inclusive training data, more reliable tools for making AI-generated predictions explainable or interpretable, and more robust regimes of algorithmic auditing and accountability to find the harmful distortions and unexpected occlusions that persist in these mirrors—even when we develop them with care, rigor, and integrity.

But there are limits—hard limits—to this path of building better, more encompassing AI mirrors. At least this is true for the data-hungry AI models that dominate the market today. An algorithmic pattern discriminator that has no body to burst with energy or ache with age, no despair or ennui to assuage, no dreams to pursue, no calls for justice to make or answer, no secrets to hold

or share, no hopes or fears to express in song or dance, no larger purpose to find in service or solidarity with others, can never know very much about who or what I am, or about who *we* are. More importantly, it knows even less—nothing at all—about what we can become, what we might make of ourselves and our lives together.

But *we* need this knowledge; and not just of ourselves but even more so of our shared human reality and potential. The future for the human family is going to be very rough going for a while. We cannot secure our shared survival, much less a future where humans flourish, without understanding our fullest potential for moral responsibility, for transformative change, and for solidarity with one other. For this reason, it is an existential necessity—the most vital task—that we not only hold onto but deepen our knowledge of who we are, and what we can do together. AI can't do this for us. So, let us build more inclusive, more reliable, and more faithful AI mirrors where we can, and use them happily for those good purposes that no better tool can serve. My own research involves many projects to help industry, government, and nonprofit organizations do just that. We must pursue these paths while staying open to more radical and sustainable possibilities for AI systems—ones built on newer and richer foundations of value.

Today's data-hungry tools are being built by powerful corporations to feast like insatiable parasites on our own words, images, and thoughts, strip away their humane roots in lived experience, and feed them back to us as hollow replacements for our own minds. AI mirrors aren't all we've got; many other types of AI can serve as scientific and commercial tools. Still, it's worth asking: could AI one day do more? Could AI support our capacities for justice in solidarity with one another, even with other planetary life and future generations? Could AI enrich, rather than replace or diminish, our own humane practices of social care, even love? Could

future AI systems call us to self-responsibility, rather than make tomorrow's hard choices for us? Could AI one day not merely reflect our intelligence, but enable our *wisdom*?

None of this is too much to hope for. Until the day that we achieve these hopes, let us be wary of refitting our homes, workplaces, courtrooms, public squares, and civic meeting spaces as endless halls of AI mirrors. We can still avoid the fate of Narcissus, captured by the dazzling, narrow light of our own reflections. We can still alter the trajectories predicted from our mirrored past. We can refuse to surrender the futures we might yet build for ourselves—and for one another.